

Patrick Flanagan

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EDUCATION

Maynooth University, Ireland

PhD Student in Astrophysics

Researching star and planet formation

Supervisor: Dr. Emma Whelan

September 2025 - present

Maynooth University, Ireland

BSc in Physics with Astrophysics

1st class honours (79.9%)

Received the Callan Award in recognition of the highest mark across the degree.

Completed modules in experimental and theoretical physics, computer science, and maths.

2021 – 2025

Patrician Secondary School, Newbridge

Leaving Certificate: 625 points

Achieved highest possible grade in final examinations.

Received H1s in English, Irish, Maths, Physics, Chemistry, Geography, and German.

2015 – 2021

PUBLICATIONS

Mauve Science Collaboration - Year 1 (2026)

“The Preliminary Mauve Science Programme: science themes identified for the first year of operations”,

RAS Instruments and Techniques, volume 5,

doi: 10.1093/rasti/rzag018.

PRESENTATIONS

ESO Writing School 2026

Spoke about my research into Herbig Ae/Be stars.

Focused on the background of Herbig stars, the methods I will be using, and the instruments I will observe with.

Garching bei München, Germany

20th-24th July 2026

Maynooth Faculty of Science and Engineering (FSE) Research Connections

Presented a poster describing our ongoing survey of Herbig Ae/Be stars.

Maynooth University, Ireland

14th May 2026

Thesis presentation

Spoke about my thesis research into planetary nebulae.

Maynooth University, Ireland

May 2025

Summer Programme for Undergraduate Research (SPUR) 22nd October 2024

Presented a poster about the history of teaching and learning physics and maths in Maynooth.

Material archived in the Russell library was used for this research. Supervised by Dr. Ciarán Mac an Bhaird.

Maynooth University, Ireland

22nd October 2024

RESEARCH EXPERIENCE

PhD student at Maynooth University

September 2025 - present

- I am part of the Mauve Science Consortium – Year 1. This provides guaranteed observing time with the Mauve Space Telescope to observe young stellar objects (YSOs) in the UV-visible wavelength range. I have worked on preparing Python tools for reducing and analysing this data. [Github]
- I received data from the Tautenberg Coudé Echelle Spectrograph (TCES) for 3 stars, which will be part of our survey with Mauve. TCES provides high-resolution optical spectra. I wrote Python tools to reduce and analyse this data.
- I used archival Transiting Exoplanet Survey Satellite (TESS) data to understand the variability of our survey targets. By building light curves on the order of 10s of days, I confirmed that daily observations with Mauve will build a light curve free from aliasing effects.
- I performed a literature review for 40 different YSO targets for the survey. The targets ranged from Herbig Ae/Be stars (our main targets) to classical T Tauri stars (CTTS). I catalogued key stellar parameters and coordinates for planning future observations.
- I applied for ESO's Fast Track Proposal channel (FTC) on the 15th of June 2026. I am PI of the proposal titled: "VLT+Mauve monitoring of Accretion-Ejection Connection in Herbig Ae/Be stars". This included researching suitable targets and verifying their visibility from the VLT, determining suitable observation and exposure times for each target using Exposure Time Calculators (ETCs), and selecting the appropriate instruments. The instruments applied for were VLT/ESPRESSO and VLT/XSHOOTER.

Final year project

2024-2025

- My 4th year thesis research used observations I took from OHP in France of a set of planetary nebulae. I planned and conducted the observations on the 0.8m and 1.52m telescopes, reduced the data, analysed spectra, and constructed temperature and density maps of our targets. I researched the methods I applied through books from the library and articles on the NASA ADS.

SPUR internship

Summer 2024

- I took part in the Summer Programme for Undergraduate Research 2024. I investigated old material present in the Russell Library and the Science Museum in the hope of discovering more about the history of teaching and learning maths, physics, and astronomy in Maynooth. I uncovered journals written by the first head of natural philosophy, Abbe Darre, and followed a timeline of the development of the curriculum through to Nicholas Callan in the 1860s.

TEACHING EXPERIENCE

Physics lab demonstrator (Maynooth University)

Demonstrated for:

- *3rd year undergraduate experiments.* Advised students on experiments, calibrated equipment, corrected reports, and managed grades with other staff. Demonstrated for cosmic ray telescope, Cavendish apparatus, interfacing, and operational amplifier experiments. From October 2025 to May 2026.
- *1st year undergraduate experiments.* Advised students on experiments, calibrated equipment, corrected reports, and managed grades with other staff. Demonstrated for lenses and basic electricity experiments. From October 2024 to May 2025, and February 2025 to May 2025
- *Leaving Certificate experiments.* Demonstrated for a wide range of leaving cert experiments provided by the physics lab in Maynooth. January 2026

Supervised:

- *3rd year undergraduate projects.* 2 different projects – the determination of the refractive index of air with changing pressure using a Michelson Interferometer and an Arduino Uno, and the rate of decay of muons using a cosmic ray telescope. From March 2026 to May 2026.

CONFERENCE PARTICIPATION

- ESO summer writing school, July 2026
- Mauve science team meeting, October 2025

SKILLS AND INTERESTS

- Reduction and analysis of astronomical images and spectra
- Development of tools for the above in Python
- Public speaking and presentation skills
- Observation planning and proposal writing
- Teaching physics
- Use of telescopes for observations
- Experienced in Microsoft Excel
- Proficient in Java and Python
- Coding experience in HTTP, SQL, C++
- Moderately proficient in German
- Enjoy learning new languages
- Keen interest in astrophysics and cosmology

AWARDS AND ACHIEVEMENTS

- Maynooth Callan Award 2025
- Maynooth University Student Excellence (MUSE) Gold Award 2025
- 6th year student of the Year 2021, and Modern Language student of the year 2021 (German)

REFEREES

Dr. Emma Whelan
PhD supervisor
Emma.whelan@mu.ie

Dr. Patrick Kavanagh
4th year thesis supervisor
Patrick.Kavanagh@mu.ie

Dr. Ciaran Mac an Bhaird
SPUR project supervisor
Ciaran.MacanBhaird@mu.ie

Dr. Neil Trappe
Maynooth University Physics Department Head
Neal.A.Trappe@mu.ie